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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	TEAM SALAMANCA
Selected Domain	DOMAIN 1
Selected SDG Track (1–17)	SDG 4 AND 8
Project / Solution Name	Futurique - Your Future. Your Skills. Your Path.
Name (Leader)	CHARUMATHI A/P SATHIA GUNALAN
Email & Phone (Leader)	scharu2772@gmail.com (011-10745483)
University Affiliation(s)	UNIVERSITI TEKNOLOGI PETRONAS, UNIVERSITI TEKNOLOGI MALAYSIA, ASIA PACIFIC UNIVERSITY & XIAMEN UNIVERSITY MALAYSIA
Number of Team Members	5 MEMBERS

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Our team is tackling the growing mismatch between rapidly advancing artificial intelligence (AI) technologies and the skills people need to remain employable in the future. As AI transforms industries, automates repetitive tasks, and creates new job opportunities, students, fresh graduates, and working adults face increasing uncertainty about their career prospects. Many struggle to identify which occupations may decline, which skills will become more valuable, and what steps they should take to remain competitive.

The challenge is not simply a lack of information, but the difficulty of turning fragmented, constantly changing labour-market information into practical, personalised career decisions. Existing career guidance often focuses on current qualifications and job openings, leaving individuals with limited insight into emerging skills, future workforce scenarios, and the most relevant reskilling opportunities. At the same time, employers and human resource teams must identify workforce skill gaps and prepare employees for technological changes, often under constraints of time, budget, and resources.

This challenge has significant economic and social consequences. Without accessible and timely guidance, individuals risk investing in outdated skills, experiencing prolonged unemployment, or being left behind by technological disruption. These risks may disproportionately affect people with limited access to career counselling, professional networks, and affordable training. Employers may also face skills shortages, reduced productivity, and higher retraining costs when their workforce cannot adapt quickly enough.

Our solution, Futurique, aims to bridge this gap by connecting labour-market trends, future career opportunities, personalised skill-gap analysis, and accessible upskilling pathways. By making career recommendations more transparent and actionable, Futurique seeks to help individuals make informed decisions, support employers in workforce planning, and promote lifelong learning. Ultimately, we aim to make the transition into an AI-driven economy more inclusive, resilient, and aligned with the United Nations Sustainable Development Goals, particularly quality education, decent work, and reduced inequalities.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our proposed application, Futurique, is an AI-powered career intelligence platform designed to help students, fresh graduates, and working adults prepare for the changing job market. By combining labour-market trends, career recommendations, and personalised upskilling pathways, Futurique aims to transform uncertain career decisions into informed and actionable plans.

The application will use real-world labour-market data and job-market trends to identify emerging career opportunities, changing skill requirements, and occupations potentially affected by AI disruption. Through AI-powered analysis, Futurique will compare users' existing skills, qualifications, and interests with the requirements of potential future roles. It will then generate explainable career compatibility scores, highlight skill gaps, and recommend relevant learning activities and accessible courses to help users progress towards their target careers.

A key feature is the Career Crash Test, which allows users to explore different future scenarios, such as moderate and rapid AI adoption, and understand how these changes could affect career opportunities and skill requirements. Futurique will also connect recommended careers and skills to relevant United Nations Sustainable Development Goals (SDGs), promoting inclusive education, decent work, and reduced inequalities.

Building on the SDC Phase 1 workshops, we intend to leverage the practical skills gained from using Replit to develop, test, and deploy our application. Replit enables us to build and refine the platform efficiently, allowing us to focus on developing functional features and improving the user experience. We also intend to explore integrating Generative AI APIs to support personalised career guidance and natural-language explanations, alongside database architecture for managing user profiles, skills, job-market information, and learning resources. Cloud hosting will help make the application accessible to users through the web.

Ultimately, Futurique aims to make future career planning more accessible, personalised, and data-informed, empowering individuals and employers to adapt confidently to an AI-driven economy.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Futurique uses a web-based architecture with the following technologies:

Frontend: React 19 and TypeScript, built with Vite 7. Tailwind CSS and Radix UI provide the interface, Recharts displays workforce charts, and TanStack Query manages API requests.

Backend: Node.js 24 with Express 5 provides REST endpoints for vacancy searches, CV extraction, job matching, and labour-market information.

Database: PostgreSQL 16 with Drizzle ORM stores public job listings, collection status, and refresh metadata.

Document processing: Python 3.13, pypdf, and python-docx extract text from PDF and DOCX CVs. Dictionary-based rules identify skills, which users can review before matching.

External services: Public job-board APIs from Ashby, Greenhouse, and Lever supply vacancies. Malaysia's data.gov.my API supplies official labour-market statistics. Learning pathways link to roadmap.sh.

API consistency: OpenAPI, Orval, and Zod maintain shared API definitions, generated frontend clients, and input validation.

System architecture: The user interacts with the React frontend, which sends requests to the Express API. The API accesses PostgreSQL, retrieves external workforce data, and invokes Python processes for vacancy collection and CV parsing. Results return to the frontend for display. Confirmed skills and selected learning paths can persist in browser local storage.

Sponsor platform usage: We used **Replit** for development. The repository includes deployment configurations for a static frontend and an autoscaled backend API. The current implementation uses rule-based matching and recommendations; **no sponsor-provided GenAI API or RunPod GPU endpoint is integrated**.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design-thinking principles by focusing on the needs of our target users: students, fresh graduates, working adults, and HR professionals who face uncertainty as AI transforms the job market.

Empathise and Define: We identified the challenges users face in understanding future job opportunities, recognising skill gaps, and finding suitable upskilling pathways. We defined our goal as making career planning more accessible, personalised, and actionable.

Ideate: We developed Futurique, an AI-powered platform that combines labour-market trends, personalised career matching, skill-gap analysis, and recommended learning pathways. We also incorporated the Career Crash Test to help users explore different AI disruption scenarios. A multilingual interface supporting English, Malay, Simplified Chinese, will make the platform more accessible to users from diverse linguistic backgrounds.

Prototype: Using Replit, we developed an initial web application to visualise the proposed features and user journeys. This allows us to refine the interface and test functionality efficiently before expanding the system.

Test and Implement: Before Phase 2, we intend to gather user feedback, improve navigation and accessibility, verify data sources, and test the accuracy and transparency of AI-generated recommendations. We will also prioritise secure data handling, reliable database integration, and cloud deployment.

Through this iterative approach, Futurique aims to evolve from an initial prototype into a technically feasible, user-friendly, and deployment-ready solution that responds to real career-planning needs.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4, Target 4.4: By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. SDG 8, Target 8.5: By 2030, achieve full and productive employment and decent work for all, including young people. SDG 4 Indicators | Institute for Statistics (UIS)
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The UN reported that 20.0% of young people globally aged 15–24 were not in employment, education or training in 2025. This highlights the scale of youth disconnection from employment and learning opportunities. It is a broad contextual baseline, not a measure of the number of people directly affected by poor career guidance or AI disruption. Goal 8 | Department of Economic and Social Affairs
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of test users who successfully identify at least one relevant skill gap and one suitable next learning activity after using Futurique. Measure this through a structured usability test with students, fresh graduates, or working

Metric Category	Your Response
	adults. Record the initial result before improvements as the prototype baseline.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 80% of test users successfully identify one relevant skill gap and one suitable learning activity using Futurique by the final prototype presentation. Test with at least 20 participants and report the actual results. This is a proposed usability target, not an established outcome.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Annual Salaries & Wages by Occupation & Sex — DOSM	Dataset (CSV)	Creative Commons Attribution 4.0 International (CC BY 4.0)
Annual Salaries & Wages by Industry & Sex — DOSM	Dataset (CSV)	Creative Commons Attribution 4.0 International (CC BY 4.0)
Monthly Principal Labour Force Statistics — DOSM	Dataset (CSV)	Creative Commons Attribution 4.0 International (CC BY 4.0)
Quarterly Skills-Related Underemployment by Age — DOSM	Dataset (CSV)	Creative Commons Attribution 4.0 International (CC BY 4.0)
Employment by MSIC Sector and Sex — DOSM	Dataset (CSV)	Creative Commons Attribution 4.0 International (CC BY 4.0)

Job-boards	Open-source job posting extractor	MIT License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes-☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



CHARUMATHI SATHIA GUNALAN

25/10/2026